Lab 08 Reflective Journal

ITAI 1371 | Lab 08: Bias-Variance Tradeoff | Team: TheBetaThetas

Brandy Gonzales

Lab 8 showed me the difference between underfitting, overfitting, and a good fit. Degree 1 was underfitting because it was a straight line that didn't waver, no matter what the data did. Degree 15 was overfitting. It tried too hard to fit in, so it pulled too hard toward each point and missed the next one. Degree 4 was a good fit. It had a smooth curve, which meant it learned the pattern, not the noise. What was challenging was understanding the difference between overfitting and a good fit, until I examined the chart carefully and asked why each line fits or doesn't.

Derrick K. Gibbs-McGlaston

In this lab, I learned more about the difference between underfitting and overfitting by looking at how different models handled the same data. Before seeing the graphs, I understood that a model could be too simple or too complicated, but the comparison made that idea easier to recognize. The degree 1 model created a nearly straight line and missed the pattern in the data. The degree 4 model followed the overall shape much better. The degree 15 model followed more of the individual changes in the data, which showed why a more complicated model is not always a better choice. I could see that the degree 15 model looked very accurate on the training examples, but its cross-validation scores were extremely negative when the training size was small. That helped me understand why training performance alone does not tell the whole story. A model needs to learn a useful pattern instead of following every detail or random change in the training data. The learning curves also helped me understand why the amount of training data matters. For degree 1, the cross-validation score improved as more examples were added, but the model still did not capture the pattern well. This showed me that more data cannot always fix a model that is too simple. For degree 15, the large gap between the training and cross-validation scores showed how differently a complex model can perform on data it has already seen compared with data used to check its performance. One part that required extra attention was interpreting the very large negative cross-validation scores for degree 15. At first, a number that far below zero can look like an error, but I learned to look at what the score says about the predictions rather than focusing only on how unusual the number looks. This lab also reminded me to read the graph labels and compare both lines before deciding how well a model is working. Overall, I learned that choosing a machine learning model is about finding a balance. A model that is too simple may miss important patterns, while one that is too complex may fit the training data but struggle on new examples.

Victor Djuma

This lab helped me understand the bias-variance tradeoff more clearly because I was able to see the difference between a model that is too simple and a model that is too complex. Before completing this lab, the meaning of bias was still a little confusing to me. I understood that machine learning models can make mistakes, but I did not fully understand how those mistakes could come from the complexity of the model itself. After working through the examples and looking at the graphs, I now understand that high bias usually means the model is too simple and underfits the data, while high variance usually means the model is too complex and overfits the training data.

One of the biggest breakthroughs for me came from comparing the polynomial models with degrees 1, 4, and 15. The degree 1 model was basically a straight line, and I could clearly see that it could not

follow the curved pattern of the sine-wave data. This helped me understand underfitting in a visual way. The model was not flexible enough to learn the real pattern. The degree 15 model was the opposite. It was very flexible and followed the training data more closely, including some of the random noise. This helped me understand why a model can learn the training data extremely well yet still perform poorly on new data. The degree 4 model helped me see what a better balance looks like because it followed the general pattern without becoming unnecessarily complicated.

The learning curves also changed the way I think about model performance. Before this lab, I might have looked mainly at one score and tried to decide whether the model was good or bad. Now I understand that comparing the training score with the cross-validation score can provide more information. When both scores are poor, it can indicate high bias or underfitting. When the training score is much better than the cross-validation score, the gap can indicate high variance or overfitting. This connects with what I learned previously about cross-validation because cross-validation helps us understand whether a model can generalize beyond the data it was trained on.

I also faced some practical challenges while completing this lab. The original notebook did not display correctly because some of the text and code were compressed together without proper line breaks. I had to use a corrected version of the notebook before I could work through the lab properly. After running the completed notebook, I also noticed a problem when I exported it to PDF. On the last page, the learning-curve graph overlapped with the Knowledge Check, and some of my answers were cut off. I corrected the print settings and exported the PDF again. Then I checked the final version to make sure both learning curves and all three Knowledge Check answers were clearly visible. This experience reminded me that completing the code is not the only part of the workflow. I also need to verify that the final output is correct and readable before submitting an assignment.

This lab also connects to concepts from earlier machine learning work. In previous labs, I learned about training and testing data, regression, classification, evaluation metrics, and cross-validation. The bias-variance tradeoff connects these ideas because it helps explain why a model may perform differently on training data and new data. A model should not only fit the training examples well; it should also generalize to data it has not seen before. This is especially important when choosing the complexity of a machine learning model.

My thinking about machine learning has changed because of this lab. I used to think that making a model more powerful or more complex would usually improve its performance. Now I understand that more complexity can actually hurt performance if the model begins learning noise instead of the general pattern. The goal is not to build the most complicated model. The goal is to find a model with the right amount of complexity for the problem.

One question that came from this lab is: How do we decide exactly how much model complexity is appropriate when working with a real-world dataset where we do not already know the true pattern? I understand that learning curves, cross-validation, and model evaluation can help, but I would like to learn more about how these tools are used together to choose the best model.

Overall, this lab gave me a deeper understanding of underfitting, overfitting, bias, variance, and model generalization. The visual comparisons and learning curves made concepts that initially seemed abstract much easier to understand. Most importantly, I learned that building a good machine learning model is about finding a balance between learning enough from the training data and still being able to perform well on new data.